

Path C - From earth to plate

The planet on your plate

The future on your plate

Idea: Of all the choices one makes, food is among the most data-rich and highest-leverage. One kg of beef emits roughly the equivalent of 60 kg of peas. California's SB1383 shows how that personal lever scales into policy; making food the most clearcut bridge from an individual choice to a systemic one.

Story

Rising action = food's footprint is larger + more variable than you think

Conflict - not just what you eat, it's also what you waste

Climax: model + SB1383 bridge

Falling action: personal playbook & policy

DIKW

Data: farm-level food life cycle assessment (LCA) + ReFED surplus

Info: emissions per kg / per protein / per calorie + where food is wasted

Knowledge: What you eat and what you waste compounds

Wisdom: Playbook + SB1383 bridge

Reverse anchor / CTA: Shift your plate and waste less - and back SB1383-style policy that scales the same choice to millions of meals

1 - Title: From earth to plate: the carbon cost of food, from your fork to state law

Beef vs peas teaser

2 - The hook - Surprising scale of food's footprint, food pantries, SB1383

Production of 1kg of beef emits same as 60kg of peas & in California, a law is already acting on this ++ see where else there are equivalent laws

Chart: The 60:1 beef vs peas contrast

Data: Poore & Nemecek (60 kg vs 1 kg CO₂ / kg figure -

-> how big is food's footprint overall?

3 - In context: zoom out - food/agriculture proportion of global emissions

Data: OWID sector

-> Within food, range is enormous. Zoom back in

4 - The spread - per-kg emissions across products

Horizontal ordered bar (beef/lamb) > cheese/pork > chicken > legumes/grains

Data: Poore & Nemecek

-> but per kg isn't a fair comparison, we eat protein not just mass

5 - Normalize per gram of protein - reframe per 100g protein and see how the ranking changes, if at all

Data: Poore & Nemecek, per-protein version

->> where in the supply chain does the damage really happen? Let's model it

6 - Supply-chain clustering

Cluster plot of the 40 products by stage profile (animal vs plant should self-separate)

Or stacked bar of stage contributions per product group showing that for beef, land-use + on-farm methane dominate - not transport.

Data: Poore & Nemecek full stage breakdown (land-use, farm, feed, processing, transport, packaging, retail)

Method: standardize features -> k-means or hierarchical k =2-5 -> silhouette score -> validate by composition

Insight: "eat local" targets the smallest slice (transport), what you eat dwarfs where it came from

-> What you eat is one lever... what you waste is another

7 - The hidden multiplier: waste

A large share of food is wasted, and wasting high-impact food multiplies its footprint

Chart: ReFED surplus by sector (farm, manufacturing, retail, foodservice, residential) with associated greenhouse gases

-> This is exactly what SB1383 targets

8 - Policy bridge : SB1383

California mandates that generators/distributors donate surplus edible food to recovery organizations rather than landfilling it (goal: rescue >20% of surplus food) connecting the individual lever to systemic. Mention how this allows my food pantry to operate and feed 270 families (~over 1000 individuals) and push about 9000 lbs of food per day

We rescued over 1 million lbs of food last year that would have otherwise ended up in landfill -> equivalent to powering a X sized village for Y amount of time

Chart: simple flow diagram maybe, surplus food -> recovery orgs -> food pantries, with landfill emissions avoided

Data; Sb 1383 framing + ReFED recovery/ greenhouse gas estimates

-> So what does this mean for one person and beyond?

9 - playbook / falling action

Concrete swaps ranked by impact, plus waste-reduction

Chart: swap X for Y with tCO₂e saved (beef -> chicken, beef -> legumes)

Data: synthesized from Poore & Nemecek

->> bigger picture...

10 - Individual vs systemic

Diet is high leverage personally, but food is one sector, SB1383 shows policy can actually multiply individual intent. Both matter.

Chart : personal lever vs policy lever, sized
Data: OWID + ReFED
-> Takeaway

11 CTA/close - one sentence big idea - callback to beef vs peas + SB1383 bridge
Beef vs peas contrast w policy bridge

12 Sources & appendix: datasets (Poore & Nemecek, ReFED), clustering metrics/more precise numbers, SB1383 reference, limitations

Datasets:

1 - Food Product Emissions (Poore & Nemecek, 2018)

[kaggle.com/datasets/amandaroseknudsen/foodproductemissions](https://www.kaggle.com/datasets/amandaroseknudsen/foodproductemissions)

This data set contains GHG emissions produced from each stage of the food product stage for 43 different food items. The emissions are calculated as a global mean and there are seven different stages. This dataset will allow us to analyze which stages produce the most emissions as well as determine which foods generate the most emissions. A possible limitation of this dataset is the number of food items not representing the entire global diet.

How do different animal vs plant products compare? How much does a diet shift save? Which stage dominates for high-impact foods? Do products separate into animal vs plant clusters

Limitations: global averages; small n=43

2- ReFED US food surplus

insights.refed.org

<https://refed.org/food-waste/resource-library>

Farm — <https://refed.org/downloads/by-sector-farm-2025.pdf> - 16.9 million tons of surplus, \$13.4 billion, 3.56 million metric tons CO₂e

Manufacturing — <https://refed.org/downloads/by-sector-manufacturing-2025.pdf> - 13.2 million tons, \$42.7 bill, 38.6 million metric tons CO₂e

Retail — <https://refed.org/downloads/by-sector-retail-2025.pdf>
4.63 million tons, \$30.3 billion, 15.1 million metric tons CO₂e

Foodservice — <https://refed.org/downloads/by-sector-foodservice-2025.pdf>
12.5 million tons, \$157 billion, 63.6 million metric tons CO₂e

Residential — <https://refed.org/downloads/by-sector-residential-2025.pdf>
23.5 million tons, \$14 bill, 100 million metric tons CO₂e

Sector	Surplus (M tons)	% of total	GHG (MMt CO ₂ e)	Value (\$B)
Residential	23.5	33.2%	100	141
Farm	16.9	23.9%	3.56	13.4

Manufacturing	13.2	18.6%	38.6	42.7
Foodservice	12.5	17.7%	63.6	157
Retail	4.63	6.5%	15.1	30.3

This data is from ReFED, a source containing food waste data and will allow us to analyze food waste by sector. Part of our project will include a discussion about California's SB1383 bill which tackles food waste so the number from this source will aid in explaining why SB1383 is important.

Which sector wastes the most by tonnage vs by CO₂e ? Where does individual vs policy action have the most leverage?

Limitations: US-only, sector definitions differ slightly compared to climate watch dataset

3 - What is SB1383

<https://oclandfills.com/sites/ocwr/files/2022-06/Repub%20Services%201383%20Organics.pdf>

As mentioned above, SB1383 is a California bill that addresses food waste so this data source will be used to provide numbers when explaining the bill.

4 - GHG Emissions (Climate watch)

https://www.climatewatchdata.org/ghg-emissions?breakBy=sector&end_year=2023§ors=total-including-lulucf&start_year=1990

This data set contains global historical emissions data across ten sectors: Agriculture, Building, Electricity/Heat, Fugitive Emissions, Manufacturing/Construction, Other Fuel Combustion, Transportation, Industrial Processes, Land Use/Forestry and Waste. This dataset will be helpful in showing how global emissions have changed over time, and also will aid us in creating a linear regression model to predict future emissions levels assuming the sector practices remain the same.

Global Historical Emissions



Data Source Climate Wa... ▾	Location World × ▾	Sectors/Subsectors 10 selected × ▾	Gases All GHG × ▾	Calculations Total ▾	Show data by Sectors ▾	Chart Type Line ▾
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CO₂e

Sector	unit	1990	1991	1992	1993	1994	1995	1996	1997	1998
Agriculture	MtCO ₂ e	5159.21	5139.69	5121.81	5090.3	5134.86	5187.05	5216.01	5141.74	5195.94
Industrial	MtCO ₂ e	1083.96	1090.16	1102.56	1110.32	1163.12	1239.44	1275	1315.6	1302.51
Land Use, Change in	MtCO ₂ e	2312.1	2312.1	2312.1	2312.54	2312.21	2317.81	2123.84	3123.19	2435.33
Waste	MtCO ₂ e	1343.22	1369.08	1439.25	1506.61	1575.39	1583.45	1587.82	1584.99	1578.57
Building	MtCO ₂ e	3232.64	3262.67	3155.61	3226.14	3141.45	3222.36	3288.41	3274.13	3119.61
Electricity	MtCO ₂ e	8649.79	8792.61	8949.72	9000.06	9101.43	9292.4	9658.58	9932	10158.5
Fugitive Emissions	MtCO ₂ e	2618.15	2618.02	2551.9	2558.47	2559.29	2634.52	2698.02	2708.64	2687.9
Manufacturing and construction	MtCO ₂ e	3967.89	3891.87	3760	3706.37	3721.82	3944.93	3827.5	3852.18	3849.95
Other Fuel Combustion	MtCO ₂ e	771.46	772.17	687.58	699.68	669.83	621.12	548.13	596.81	573.27
Transport	MtCO ₂ e	4726.28	4764.57	4872.52	4915.56	5013.54	5150.94	5360.88	5438.93	5565.35
Data source	Climate Watch									
Data for	Location:	Industrial	Land Use	Land-Use	Waste	Building	Electricity	Fugitive E	Manufact	Other Fuel

Data spans through to 2023